

Credit Card Fraud Detection using Machine Learning

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Project Overview

This project aims to detect fraudulent credit card transactions using Machine Learning. The model is trained on a dataset containing transaction details and predicts whether a transaction is genuine or fraudulent.

Objective

The main objective of this project is to identify fraudulent transactions and help financial institutions reduce losses caused by credit card fraud.

Dataset

Dataset: creditcard.csv

Features:

- Time
- Amount
- V1 to V28 (anonymized transaction features)

Target Variable:

- Class = 0 (Genuine Transaction)
- Class = 1 (Fraudulent Transaction)

Technologies Used

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn

Project Workflow

1. Data Collection

The credit card transaction dataset was loaded using Pandas.

2. Data Exploration

- Displayed first few rows of data
- Checked dataset shape
- Verified data types
- Checked for missing values

3. Data Visualization

Visualized the distribution of genuine and fraudulent transactions using count plots.

4. Data Preprocessing

- Separated features and target variable
- Split dataset into training and testing sets

5. Model Training

A Random Forest Classifier was used to train the model.

6. Model Evaluation

The model performance was evaluated using:

- Accuracy Score
- Confusion Matrix
- Classification Report

Results

The model successfully identified fraudulent transactions with high accuracy.

Future Enhancements

- Hyperparameter tuning
- Deep Learning models
- Real-time fraud detection system
- Web application deployment using Streamlit

Author

Project developed as part of Machine Learning internship practice.